

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: d92681e9 • Generated: 14:22:54.085Z

Natural Query:	Use both images to identify built-up regions.
Classified Task:	OPTICAL_SAR_ANALYSIS
Calibrated Confidence:	92% (Multi-Sensor Cross-Modal Concordance)
Specialist Models:	satquery-optical-sar-fusion-v1
Total Latency:	425.8 ms

GROUNDING AI INTERPRETATION

Optical-SAR synergistic analysis identified 15.44% built-up infrastructure. SAR microwave backscatter effectively validated structural footprints while bypassing optical cloud/atmospheric shadows.

INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
sample_optical_cloud	OPTICAL	512x512	3	None (Benchmark)
sample_sar_backscatt	SAR	512x512	1	None (Benchmark)

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (m)
1	Query Ingestion	Received natural language query: 'Use both images to identify built-up regions.' with 2 uploaded image(s).	0.0
2	Input Validation	Verified 2 image file(s). Format and size limits satisfied.	0.0
3	Raster Metadata Extraction: sample_optical_cloudy.png	Extracted dimensions (512x512), 3 band(s). Format: Standard Benchmark Image (PNG/JPEG)	73.5
4	Raster Metadata Extraction: sample_sar_backscatter.png	Extracted dimensions (512x512), 1 band(s). Format: Standard Benchmark Image (PNG/JPEG)	24.6
5	Modality Identification	Detected sensor modalities: ['OPTICAL', 'SAR'].	0.0
6	Query Classification → OPTICAL_SAR_ANALYSIS	Dual-sensor pair (Optical + SAR) detected -> Routed to Cross-Modal Feature Fusion Specialist.	0.0
7	Model Dispatch	Dispatched task to specialist: 'satquery-optical-sar-fusion-v1' on device 'cpu'.	0.0
8	Specialist Execution Completed	Inference finalized in 327.2ms. Evidence artifacts compiled.	327.3
9	Result Fusion & Calibration	Derived grounded response with calibrated confidence 92% (Multi-Sensor Cross-Modal Concordance).	0.0

